INCH-POUND

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w/AMENDMENT 2
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SUPERSEDING
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w/AMENDMENT 1
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DETAIL SPECIFICATION

MAGNESIUM POWDER - FOR USE IN AMMUNITION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers powdered magnesium suitable for use in ammunition.

1.2 Classification. Magnesium powder is of the following types, grades, and granulations:

1.2.1 Type (see 6.4).

Type I - Flaked and/or chipped.

Type II - Oblong chip with rounded edges.

Type III - Granular or spheroidal (atomized).

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, Attn: RDAR-EIQ-SE, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2.2 Granulations. The granulations are given in Table I.

TABLE I. Nominal granulations.

Granulation No.	Metric Nominal mesh size	U.S. Nominal mesh size
1	425 µm to 180 µm	40 to 80
2	425 µm to 180 µm	40 to 80 (alternate)
3	300 µm to 150 µm	50 to 100
4	300 µm to 150 µm	50 to 100 (Army)
5	300 µm to 125 µm	50 to 120
6	180 µm to 125 µm	80 to 120
7	thru 150 µm	thru 100
8	125 µm to 75 µm	120 to 200
9	thru 125 µm	thru 120
10	thru 75 µm	thru 200
11	180 µm to 75 µm	80 to 200
12	125 µm to 75 µm	120 to 200 (Army)
13	850 µm to 300 µm	20 to 50
14	300 µm to 150 µm	50 to 100
15	150 µm to 75 µm	100 to 200
16	75 µm to 45 µm	200 to 325 (see 6.3)
17	300 µm to 150 µm	50 to 100
18	600 µm to 300 µm	30 to 50

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/ or from the Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM International

ASTM B92/B92M	-	Standard Specification for Unalloyed Magnesium Ingot and Stick for Remelting [Metric]
ASTM B215	-	Standard Practices for Sampling Metal Powders
ASTM B329	-	Standard Test Method for Apparent Density of Metal Powders and Compounds Using the Scott Volumeter
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Copies of ASTM standards are available online at https://www.astm.org or from American Society for Testing and Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Material. Magnesium powder shall be manufactured from magnesium ingot or stick that meets the requirements of ASTM B92M.

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3.3 Form.

3.3.1 Form - Type I. Type I magnesium powder shall consist of shavings, turnings, flakes, or any combination of these which meet the necessary granulation requirements shown in Figure I.

3.3.2 Form - Type II. Type II magnesium powder shall consist of oblong chip-like particles with rounded edges which meet the necessary granulation requirements shown in Figure I.

3.3.3 Form - Type III. Type III magnesium powder shall consist of granular or spheroidal particles (atomized) which meet the necessary granulation requirements shown in Figure I.

3.4 Chemical composition. Magnesium powder shall conform to the requirements shown in Table II.

3.4.1 Magnesium. The magnesium shall conform to the chemical and physical requirements as shown in Table II.

3.4.2 Volatile material at 105° C. The volatile material at 105° C shall conform to the chemical and physical requirements as shown in Table II.

3.4.3 Oil and grease. The oil and grease shall conform to the chemical and physical requirements as shown in Table II.

3.4.4 Total iron. The total iron shall conform to the chemical and physical requirements as shown in Table II.

3.4.5 Carbides. The carbides shall conform to the chemical and physical requirements as shown in Table II.

TABLE II. Chemical composition.

Requirement	Type I Powder	Type II Powder	Type III Powder
Magnesium, % min	96 1/	96 1/	98
Volatile material at 105 °C, % max	0.1	0.1	0.1
Oil and grease, % max	0.02	0.02	0.02
Total Fe, % max	0.22	0.22	0.22
Carbides, % max	0.004	0.004	0.004
1/ Granulations 13 & 18 shall be 98 % min.			

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3.5 Granulation. Magnesium powder shall be of the fineness shown in Figure 1 when determined as specified in 4.5.4. All percentages shall be by weight using US Standard Sieves conforming to ASTM E11. The powder shall pass through the required sieves readily without balling or the particles clinging together. The granulations are given in Figure 1.

Granulation 1	
US Sieve	% Retained
On 30	0
On 40	0 - 10
On 50	25 - 45
On 70	30 - 50
On 80	15 - 35
Thru 80	0 - 10

Granulation 2	
US Sieve	% Retained
On 30	0
On 40	0 - 7
On 50	25 - 45
On 70	30 - 50
On 80	10 - 30
Thru 80	5 - 15

Granulation 3	
US Sieve	% Retained
On 30	0
On 50	0 - 10
On 70	45 - 60
On 80	15 - 30
On 100	15 - 30
Thru 100	0 - 15

Granulation 4	
US Sieve	% Retained
On 20	0
On 50	0 - 12
On 100	76 - 100
Thru 100	0 - 12

Granulation 5	
US Sieve	% Retained
On 40	0
On 50	0 - 10
On 120	80 - 100
Thru 120	0 - 10

Granulation 6	
US Sieve	% Retained
On 70	0
On 80	0 - 10
On 120	80 - 100
Thru 120	0 - 15

Granulation 7	
US Sieve	% Retained
On 100	0 - 2
Thru 100	98 - 100

Granulation 8	
US Sieve	% Retained
On 60	0
On 80	0 - 1
On 120	0 - 15
On 200	75 - 100
Thru 200	0 - 10

Granulation 9	
US Sieve	% Retained
On 120	0 - 2
Thru 120	98 - 100

Granulation 10	
US Sieve	% Retained
On 120	0
On 200	0 - 10
Thru 200	90 - 100

Granulation 11	
US Sieve	% Retained
On 25	0
On 80	0 - 30
On 120	30 - 50
On 200	25 - 45
Thru 200	0 - 25

Granulation 12	
US Sieve	% Retained
On 100	0
On 120	0 - 3
On 200	15 - 40
Thru 200	57 - 85

Granulation 13	
US Sieve	% Retained
On 6	0
On 8	0 - 1
On 12	0 - 5
On 20	10 - 50
On 50	50 - 80
Thru 50	0 - 5

Granulation 14	
US Sieve	% Retained
On 30	0
On 50	0 - 10
On 70	45 - 60
On 80	15 - 30
On 100	15 - 30
Thru 100	0 - 15

Granulation 15	
US Sieve	% Retained
On 50	0
On 100	0 - 15
On 200	Balance
Thru 200	0 - 15

Granulation 16	
US Sieve	% Retained
On 100	0
On 200	0 - 40
On 325	Balance
Thru 325	0 - 50

Granulation 17	
US Sieve	% Retained
On 30	0
On 50	0 - 15
On 100	Balance
Thru 100	0 - 15

Granulation 18	
US Sieve	% Retained
On 16	0 - 1
On 30	0 - 10
On 40	20 - 60
On 50	20 - 60
On 70	0 - 10
Thru 70	0 - 1

FIGURE 1. Granulations requirements.

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3.6 Apparent density. The apparent density of the magnesium powder when determined shall be shown in Table III and ASTM B329.

TABLE III. Apparent density. 1/

Granulation Number	Density					
	Type I		Type II		Type III	
	Min 2/	Max 2/	Min 2/	Max 2/	Min 2/	Max 2/
1	0.55	0.65	0.55	0.65	0.95	1.00
2	0.55	0.65	0.55	0.65	0.92	1.00
3	0.65	0.75	0.65	0.75	0.92	1.05
4	0.45	0.625	0.45	0.625	0.92	1.05
11	0.45	-	0.45	-	-	-
12	0.45	-	0.45	-	-	-
13	0.70	-	0.70	-	-	-
14	0.65	0.75	0.65	0.75	0.92	1.05
15	0.62	-	0.62	-	0.90	-
16	0.57	-	0.62	-	0.57	-
17	0.75	-	0.75	-	0.90	-

1/ Granulations 5-10 and 18 do not have an apparent density requirement.
2/ Density in grams per milliliter (g/mL).

3.7 Workmanship. The metal powder shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter, or other defects that would render the metal powder unsuitable for intended use.

3.8 Process verification.

3.8.1 Types I and II. Type I and II magnesium powder shall be manufactured by grinding, stamping, turning, or any other combination of these methods which will yield a suitable product.

3.8.2 Types III. Type III magnesium powder shall be manufactured by the atomizing process.

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4. VERIFICATION

TABLE IV. Requirement/verification cross reference matrix.

Method of verification		Classes of verification						
1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		A – First article B – Conformance						
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X	X	X	X	X		4.2
3.1.2	Conformance	X	X	X	X		X	4.3
3.2	Material	X		X	X	X	X	4.5.1
3.3.1	Form - Type I			X		X	X	4.5.2
3.3.2	Form - Type II			X		X	X	4.5.2
3.3.3	Form - Type III			X		X	X	4.5.2
3.4	Chemical composition				X	X	X	4.5.3
3.4.1	Magnesium				X	X	X	4.5.3.1
3.4.2	Volatile material at 105°C				X	X	X	4.5.3.2
3.4.3	Oil and grease				X	X	X	4.5.3.3
3.4.4	Total Fe				X	X	X	4.5.3.4
3.4.5	Carbides				X	X	X	4.5.3.5
3.5	Granulation				X	X	X	4.5.4
3.6	Apparent density				X	X	X	4.5.5
3.7	Workmanship			X		X	X	4.5.6
3.8	Process verification		X			X	X	4.5.7

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article (see paragraph 4.2)

b. Conformance (see paragraph 4.3)

4.2 First article inspection. When specified, a sample shall be subject to first article verification in accordance to Table V.

4.2.1 First article quantity. The first article sample shall be a minimum of 10 kilograms of metal powder from the first production lot produced by the contractor using the same production equipment, process procedures, and material supply sources as will be used in regular production.

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4.2.2 Inspections to be performed. The first article sample shall be subjected to inspection and tests specified in Table V and 4.4.

4.2.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

TABLE V. First article acceptance inspection.

Classification	Metal Powder First Article Acceptance Test	Requirement Paragraph	Inspection Method Reference
Critical	None defined		
Major			
101	Material	3.2	4.5.1
102	Form	3.3	4.5.2
103	Chemical composition	3.4	4.5.3
104	Granulation	3.5	4.5.4
105	Apparent density	3.6	4.5.5
Minor			
201	Workmanship	3.7	4.5.6
202	Process verification	3.8	4.5.7

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916. A lot shall consist of no more than one type of powder and no more than one granulation.

4.3.2 Classification of characteristics. Conformance inspection shall be performed in accordance with inspection provisions set forth herein. The characteristics shown in 3.2 through 3.8 when tested in accordance with 4.5 shall constitute minimum inspections to be performed.

4.3.3 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraph 4.3.3.1, the definitions of critical, major and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.

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4.3.3.1	Metal powder.	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None defined			
Major				
101	Material	4.4.1	3.2	4.5.1
102	Form	4.4.1	3.3	4.5.2
103	Chemical composition	4.4.1	3.4	4.5.3
104	Granulation	4.4.2	3.5	4.5.4
105	Apparent density	4.4.1	3.6	4.5.5
Minor				
201	Workmanship	4.4.2	3.7	4.5.6
202	Process verification	4.4.1	3.8	4.5.7

4.4 Metal powder sampling.

4.4.1 Sampling requirements. Two independent representative samples, minimum two kilograms, shall be obtained from each lot of magnesium powder using ASTM E300 or ASTM B215. Unless otherwise specified, sampling shall be conducted in accordance with Table VI. When lots comprise three or less containers, all containers in the lot shall be selected. Before sampling, the contents of each container (selected for sampling) shall be thoroughly blended for 10 minutes by rolling and turning end for end. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

TABLE VI. Sampling.

Containers in a Lot	Number of Samples
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
20,001 to 35,000	15
over 35,000	20

4.4.2 Sampling for granulation and workmanship. Each container shall be rolled and tumbled for ten minutes or until the metal powder is completely mixed in the container. Then a sample of minimum 227 grams shall be removed from the sample with a sampling thief. The granulation and particle size shall be determined on the sample thus obtained. Each container shall be individually tested. If any sample fails to comply with the requirements of Figure 1, the container shall be rejected.

4.4.2.1 Alternate sampling for granulation and workmanship. The composite sample prepared in accordance with 4.4.1 may be used in lieu of the samples required in 4.4.2. If the

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composite sample fails to comply with the requirements of Figure 1, the lot shall be rejected, or each container may be sampled in accordance with 4.4.2 for acceptance determination and if any container fails to comply with the requirements of Figure 1, the container shall be rejected.

4.5 Test methods and procedures. Unless otherwise specified, all chemicals and reagents shall be ACS Grade or reagent grade chemicals.

4.5.1 Material inspection. Material shall be inspected in accordance with the "Methods of Analysis" section of ASTM B92M.

4.5.2 Form. Thoroughly mix the sample until it is homogeneous and withdraw a 10 gram sample for examination, then place the sample into a container, cover, and tumble the container for five minutes. With a scoop, remove a ¼ gram sample from the container and sprinkle the sample on a microscopic slide and examine under a 20 to 100 power microscope. (See 6.4).

4.5.3 Chemical composition.

4.5.3.1 Magnesium.

4.5.3.1.1 Magnesium (preferred method).

4.5.3.1.1.1 Reagents.

a. Chromic acid solution (approximately 0.1 N as acid). Dissolve 10.00 grams of chromium trioxide (CrO_3) in distilled water and dilute to 1000 milliliters (mL).
b. Magnesium standard solution – Dissolve 1.8 grams of pure magnesium in 125 milliliters of 1:4 hydrochloric acid. Dilute to 1000 milliliters in a volumetric flask with distilled water. (One milliliter contains approximately 1.8 milligrams of magnesium). W_A = Actual weight of magnesium used in grams.
c. Ethylenediaminetetraacetic acid (EDTA) reagent solution – Dissolve 40.0 grams of disodium dihydrogen ethylenediaminetetraacetate dihydrate in distilled water and dilute with distilled water to 1000 milliliters in a volumetric flask.
d. Buffer solution (approximately pH 10) – Dissolve 65.5 grams of reagent grade ammonium chloride in 300 milliliters of distilled water. Add 570 milliliters of concentrated (28-30 % w/w) ammonium hydroxide and dilute to 1000 milliliters with distilled water.
e. Eriochrome Black T indicator solution – Dissolve 0.8 grams of Eriochrome Black T indicator solution in a mixture of 40 milliliters methanol and 60 milliliters of triethanolamine.

4.5.3.1.1.2 Standardization.

a. Pipette 50.00 milliliters of magnesium standard solution into a 600 milliliter flask. Dilute to about 300 milliliters with distilled water.

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b. Add 20 milliliters of buffer solution and 6-10 drops of Eriochrome Black T indicator solution. Titrate with EDTA reagent solution to a color change of red to blue against a reflected light background. There should be no red color remaining. V_A = Volume of EDTA reagent solution used in milliliters.

$$X = \text{Milligrams of magnesium per milliliter EDTA} = 50 \cdot \frac{W_A}{V_A}$$

Where:

W_A = Actual weight of magnesium used in grams.
V_A = Volume of EDTA reagent solution used in milliliters.

4.5.3.1.1.3 Procedure.

a. Weigh approximately 1 gram (± 0.2 mg) of sample into a 250 milliliter beaker. W_B = weight of actual sample used.
b. Add 50 milliliters of 0.1 N chromic acid to the sample and allow to stand for 15 minutes, swirling the beaker every five minutes. Transfer the sample to a 30 milliliter, fine porosity sintered glass crucible and wash the sample with 25 milliliter portions of distilled water 6 times. The last washing should be colorless.
c. Wash the sample back into the original beaker with distilled water.
d. Treat the crucible with two 10 milliliter portions of 10 % hydrochloric acid to dissolve any metal remaining in it, returning the washings into the beaker with distilled water. Cover the beaker with a watch glass.
e. Add 10 milliliters of concentrated hydrochloric acid in small amounts to the sample. Gradually heat the beaker to 100°C maximum to dissolve any remaining metal.
f. Cool and transfer the solution to a 500 milliliter volumetric flask.
g. Dilute the solution to the 500 milliliter mark with distilled water and mix.
h. Pipette 50.00 milliliters of this solution into a 600 milliliter beaker.
i. Dilute the sample solution to 300 milliliters with distilled water.
j. Add 20 milliliters of buffer solution and 6-10 drops of Eriochrome Black T indicator solution to the sample solution. Titrate with EDTA reagent solution to a color change of red to blue against a reflected light background. There should be no red color remaining. V_B = Volume of EDTA reagent solution used in milliliters.

$$\% \text{ metallic magnesium} = V_B \cdot \frac{X}{W_B}$$

Where:

X = Milligrams of magnesium per milliliter EDTA reagent solution.
V_B = Volume of EDTA reagent solution used in milliliters.

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W_B = weight of actual sample used.

4.5.3.1.2 Magnesium (eudiometer method) (alternate method). The magnesium shall be determined in accordance to method 412.1 of Standard MIL-STD-1234.

4.5.3.2 Volatile material at 105ºC.

a. Approximately 5 grams of the sample shall be weighed into a tared dish and heated in an oven for 3 hours at 105ºC.
b. Remove the sample from the oven and place into a dessicator at room temperature to cool.
c. Reweigh sample and the loss in weight shall be calculated as percent volatile material at 105ºC.

$$\text{Percent volatiles} = \frac{A \cdot 100}{B}$$

Where:

A = loss in weight, (gm).
B = weight of sample, (gm).

4.5.3.3 Oil and grease.

a. A weighed portion of approximately 20 grams of the sample shall be extracted with diethyl ether or hexane in a Soxhlet or similar extractor for minimum 6 hours using a tared flask.
b. The diethyl ether shall be evaporated by steam bath when extraction is completed in the flask.
c. The contents shall be dried at 90ºC, to constant weight, cooled in a desiccator, and weighed. A blank shall be run at the same time.
d. The weight of the residue shall be calculated to percent oil and grease, as follows:

$$\text{Percent oil and grease} = \frac{100 \cdot (A-B)}{W}$$

Where:

A = weight of ether extract.
B = weight of blank.
W = weight of sample.

4.5.3.4 Total Fe (ICP method).

4.5.3.4.1 Reagents. Ultra high purity acids shall be used for sample dissolution.

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It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Alternately, stock solutions may be prepared in-house from high purity metals or their salts. The maximum recommended shelf life is one year from date of initial preparation.

a. Certified Fe stock solution shall contain 100.00 µg/mL Fe in 5.00 % 1:1 HNO_3 (v/v).
b. Certified Mg stock solution shall contain 20,000.00 µg/mL Mg in 0.10 % 1:1 HNO_3 (v/v)
c. Nitric Acid (HNO_3), concentrated, 67 – 70 %.
d. Distilled water.

4.5.3.4.2 Procedure.

4.5.3.4.2.1 Sample Dissolution.

a. Weigh one gram of sample, as "M", to the nearest 1 milligram and transfer into 400 milliliter beaker.
b. Add 10 milliliters of distilled water to the beaker followed by 2 milliliters of 1:1 HNO_3. (Slowly add acid and allow reaction to subside.)
c. Gradually add 16 milliliters of 1:1 HNO_3 to the sample solution, approximately 1 milliliter at a time. Reaction may become overly vigorous by adding reagent too quickly.
d. Cover the beaker with watch glass and heat on hot plate at approximately 180ºC to completely decompose sample.
e. Allow sample to cool to ambient temperature and transfer into 100 milliliter volumetric flask.
f. Rinse lower surface of watch glass and inner walls of beaker with distilled water into the flask with the sample solution.
g. Dilute the sample solution with distilled water to 100 milliliters.

4.5.3.4.2.2 Preparation of calibration curve.

a. Dispense 50 milliliters of Mg stock solution into each of two 100 milliliter volumetric flasks.
b. Add stepwise 20.0 milliliters, and 30.0 milliliters Fe stock solution separately to the two volumetric flasks and dilute with distilled water to 100 milliliters. These solutions will represent the mid and high standards respectively.
c. Prepare blank by pipetting 50 milliliters of Mg stock solution into a third 100 milliliter volumetric flask and dilute with distilled water to the 100 milliliter mark.

Mid standard shall contain: 20 µg/mL Fe and 10,000 µg/mL Mg.
High standard shall contain: 30 µg/mL Fe and 10,000 µg/mL Mg.
Blank shall contain 10,000 µg/mL Mg.

d. Aspirate standard solutions into ICP spectrometer in order of increasing concentration.

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Take emission measurements for iron and generate a calibration curve utilizing the instrument's software. Fe emission intensity shall be measured using radial view at the 259.940 nm wavelength. The calibration shall be forced through the origin. Repeat the calibration if the correlation determination (R^2) for Fe is < 0.9999.

4.5.3.4.2.3 Sample Analysis.

a. Aspirate sample solution into ICP spectrometer and record Fe emission measurement. Software generated concentrations for Fe shall be inserted into the formula below as A_1.
b. Aspirate blank solution containing 10,000 µg/mL Mg and insert into formula below as A_2.

4.5.3.4.2.4 Sample Calculation.

$$X = \frac{100 \cdot (A_1 - A_2)}{M}$$

Where:

X = content of the iron in sample [%(mass fraction)].
A_1 = amount of the iron detected in the sample (g).
A_2 = amount of iron detected in the blank test solution (g).
M = amount of sample weighed out (g).

4.5.3.5 Carbides.

4.5.3.5.1 Gas chromatography with flame ionization detection (GC/FID).

4.5.3.5.1.1 Overview. The following procedure is used to determine the amount of magnesium carbide (acetylide) in magnesium. Sample is dissolved in 5 % sulfuric acid and acetylene thus produced is measured by GC/FID. The method can be extended to measure other types of carbides present provided that appropriate standards are available and the instrument method is modified appropriately.

4.5.3.5.1.2 Reagents.

a. Sulfuric acid (H_2SO_4), concentrated 95 – 98 % wt, ACS reagent grade.
b. Distilled water.
c. Sulfuric acid, 5 % wt aqueous solution, made by dissolving 30 milliliters of concentrated sulfuric acid in enough distilled water to make 1000 milliliters of solution. Scale the amounts proportionally to make more or less reagent depending on the number or samples to be run. Quantity: 100 milliliters per sample and 50 milliliters per standard.

4.5.3.5.1.3 Gas standard.

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a. Compressed gas cylinder containing 1 mole % each C_1-C_2 hydrocarbons, i.e. methane, acetylene, ethylene, and ethane, e.g. Sigma 23462 or equivalent. Matching regulator required; see supplies and equipment (4.5.3.5.1.4).

4.5.3.5.1.4 Supplies and equipment.

a. Gas chromatograph (GC) column: Agilent J&W CarbonPLOT 60 m/0.32 mm/1.5 micometers (length/diameter/film thickness) or equivalent capable of resolving C_1-C_2 hydrocarbons in less than 10 minutes. Quantity: 1.

b. Round-bottom-3-neck flask, 500 milliliter, 24/40 ground-glass ports, e.g. Kontes 606000-1024 or eq. Quantity: 1.

c. 24/40 ground-glass barbed adapter, e.g. Kontes 182900-2440 or eq. Quantity: 2.

d. Addition funnel with stopcock, >50 milliliter, 24/40 ground glass joint, e.g. Ace Glass 7257-50 or eq. Quantity: 1.

e. Valve, 3-way, glass stopcock, e.g. Kontes 843900-0002 or eq. Quantity: 1.

f. Rubber stopper/septum fitting barbed adapter, above, e.g. Fisher 20190 or eq. Quantity: 1 per sample run.

g. Gas regulator, single stage CGA-165 with syringe adapter, e.g. Sigma 501395 or eq. Quantity: 1.

h. Powder funnel, preferably metal, with 5 centimeter long stem fitting 24/40 port or eq. made from aluminum foil. Quantity: 1.

i. Syringe, 0.1 milliliter, gas-tight with locking valve and 22 or 23 gauge, 2 inch needle, e.g. SGE 100R-V-GT or eq. Quantity: 1, spares recommended.

j. Syringe, 25 milliliter, gas-tight with locking valve and Luer lock connection, e.g. SGE 25MR-VLLMA-GT or eq. Quantity: 1.

k. Needle, 22 or 23 gauge, 2 inch length, fitting Luer lock connection, e.g. SGE 039802 or eq. Quantity: 1, spares recommended.

l. Silicone grease, high vacuum. Quantity: approximately 5 grams.

m. Support stand with clamp assortment. Quantity: 1, clamps as needed.

n. Tubing, thick-wall rubber, 1/4 inch ID. Quantity: approximately 1 foot.

o. Tubing, thick-wall rubber, 1/8 inch ID. Quantity: approximately 1 foot.

p. Union, barbed, reducing 1/4 in to 1/8 inch. Quantity: 1.

q. Volumetric glassware, assorted. Quantity: as needed.

r. GC with flame ionization detector (FID) and computer/integrator. Compressed air, hydrogen, and helium supply as recommended by GC manufacturer. Quantity: 1.

s. Vacuum pump or other vacuum source, 5 kiloPascal ultimate vacuum or better in less than 5 minutes, e.g. Sartorius 16615 or eq. Quantity: 1.

t. Laboratory balance, ≥ 200 gram capacity, 0.001 gram resolution or better. Quantity: 1.

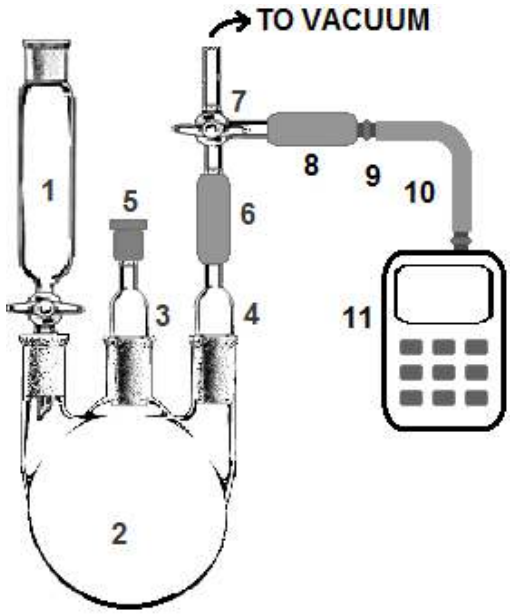
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u. Manometer, digital, range: 0 to ≥200 kiloPascal, resolution: 0.2 kiloPascal or better, e.g. Omega HHP4230 or eq. Quantity: 1.

v. Room thermometer. Quantity: 1.

w. Barometer, digital or analog, 0.2 kiloPascal resolution or better. Quantity: 1.

4.5.3.5.1.5 Equipment setup. Set up the GC. Install and condition capillary column as required by manufacturer. The following analysis conditions are recommended: oven maintained at 140°C for a 5 minute analysis time, injector at 125°C flowing helium at 2 mL/min with a 20-fold split, FID at 250°C and maximum sensitivity setting and burning a mixture of hydrogen and air, fed at 40 mL/min 400 mL/min, respectively. Assemble reaction apparatus as illustrated in Figure 2. Lubricate with silicone grease all moving parts and all joints and connections; remove excess grease. Support assembly with clamps and stand. Test gas-tightness of the assembled glassware. Evacuate the vessel and manometer connection using vacuum pump. Turn 3-way valve to exclude the vacuum pump and connect manometer and vessel. Observed pressure reading should be less than 10 kiloPascal and should not increase faster than 2 kPa/minute. Check connections and eliminate leaks.



1. Addition funnel with stopcock, >50 mL, 24/40 ground glass joint,
2. Round-bottom-3-neck flask, 500 mL, 24/40 ground-glass ports,
3. 24/40 ground-glass barbed adapter,
4. 24/40 ground-glass barbed adapter,
5. Rubber stopper/septum,
6. Tubing, thick-wall rubber, 1/4 inch ID,
7. Valve, 3-way, glass stopcock,
8. Tubing, thick-wall rubber, 1/4 inch ID,
9. Union, barbed, reducing 1/4 in to 1/8 inch,
10. Tubing, thick-wall rubber, 1/8 inch ID,
11. Manometer, digital, 0 to ≥30 psi.

FIGURE 2. Reaction apparatus set-up & connections.

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4.5.3.5.1.6 Blank Procedure.

a. Add 50.0 milliliters of 5 % wt sulfuric acid reagent to the addition funnel.

b. Evacuate reaction vessel, then turn 3-way valve to exclusively connect the flask and manometer.

c. Open addition funnel to allow acid to dispense; keep stopcock open long enough for air to enter flask and internal pressure to reach atmospheric.

d. Wait 5 minutes for the flask to warm to room temperature and open and close the stopcock to equilibrate pressure.

e. Puncture the rubber septum with the 0.1 milliliters syringe.

f. With the syringe valve open and needle inserted through the septum, slowly pump the plunger 5-10 times to fill and empty the syringe.

g. Fill the syringe with 100 microliters of gas from the flask, close the syringe valve and remove the needle from the septum.

h. When ready to make injection into GC, depress the plunger of the syringe to 80 microliters and open the syringe valve.

i. Depress the plunger further to 50 microliters and quickly inject the syringe contents onto the GC and run the sample.

j. Once GC run is complete, take a second sample from the flask and inject the same way (4.5.3.5.1.6 e. through i.).

k. Identify and integrate the acetylene peak in each chromatogram. Under the recommended conditions, C_1-C_2 peaks should elute between 3 and 4 minutes in order of increasing molecular weight, however the blank may not show any signal above noise. Acetylene is the second fastest-eluting peak. Average the acetylene peak areas and label this value “A”. Keep the sulfuric acid in the flask for preparation of calibration standard.

4.5.3.5.1.7 Calibration standard procedure.

a. Attach regulator to reference gas cylinder. Check for leaks and proper operation.

b. Open main cylinder valve. Lower the outlet pressure by turning the regulator knob completely counter-clockwise. The gauge of a single stage regulator displays the gas pressure inside the cylinder, not the outlet pressure.

c. Attach needle to 25 milliliter syringe, open syringe valve and insert needle through regulator’s syringe adapter.

d. Slowly turn the regulator knob clockwise until the plunger begins to move and syringe begins to fill. Fill and empty the syringe 5 times, then fill with 25 milliliters of reference gas, close syringe valve and withdraw the needle.

e. Evacuate flask headspace using the vacuum pump and turn 3-way valve to exclusively connect flask and manometer.

f. Record the atmospheric pressure and label it “B”.

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g. Depress 25 milliliter syringe plunger to 20 milliliters and open the valve, depress plunger further to a volume between 10 and 12 milliliters, note the exact volume "C" to nearest 0.2 milliliter, and insert needle through reaction flask septum and inject syringe contents. Then, withdraw the needle from the reaction flask septum.

h. Open addition funnel stopcock to let flask fill with air.

i. Close stopcock, allow 10 minutes for flask to warm to room temperature, then open and close the valve to equilibrate pressure.

j. Withdraw a 100 microliter sample from the flask and inject 50 microliters of it onto the GC in the same manner as the blank (4.5.3.5.1.6 e. through i.). Repeat twice more.

k. Identify and integrate the acetylene peak in each chromatogram. Average the acetylene peak areas and record this value as "D".

l. Disassemble reaction vessel and dispose of the sulfuric acid. Rinse the flask with distilled water, then with acetone and air dry completely.

4.5.3.5.1.8 Sample analysis.

a. Insert the two 24/40 barbed adapters into the flask, but not the addition funnel.

b. Pre-weigh powder funnel and weighing paper.

c. Weigh (to the nearest 0.001 gram) a 0.500 ± 0.020 gram aliquot of sample magnesium.

d. Transfer sample into the flask using the powder funnel, making sure no powder gets on the ground-glass joint.

e. Re-weigh the powder funnel and weighing paper to determine the weight "E" of sample transferred to the flask.

f. Completely assemble the reaction apparatus, and transfer 50.0 milliliters of 5 % wt sulfuric acid to the addition funnel.

g. Evacuate the flask using vacuum pump.

h. Turn 3-way valve to connect flask and manometer exclusively.

i. Open addition funnel stopcock to dispense acid into reaction flask; shut the stopcock immediately after all the acid has dispensed. Wait 15 minutes after bubbling completely stops in the reaction flask.

j. Record the flask internal pressure as "F", and the atmospheric pressure "G".

k. Withdraw two 100 microliter samples from the flask and inject 50 microliters of each onto the GC using the same technique as the blank (4.5.3.5.1.6 e. through i.).

l. Average acetylene peak areas and record the value as "H".

m. Rinse the flask with distilled water, then with acetone and air dry completely.

n. Analyze a duplicate aliquot of magnesium the same way (4.5.3.5.1.8 a. through m.).

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4.5.3.5.1.9 Sample calculation. Evaluate the expression below for each of the two aliquots and average the results to calculate carbide content in terms of magnesium carbide (acetylide). Report the average of the two aliquots.

$$\% \text{ wt } MgC_2 = \frac{0.581 \cdot B \cdot C \cdot F \cdot y \cdot (H - A)}{E \cdot G \cdot T \cdot (D - A)}$$

Where:

A = Average acetylene peak areas. (See 4.5.3.5.1.6, k.).
B = Atmospheric pressure, (bar). (See 4.5.3.5.1.7, f.).
C = Exact volume, (mL). (See 4.5.3.5.1.7, g.).
D = Average acetylene peak areas, standard. (See 4.5.3.5.1.7, k.).
E = Weight, (grams). (See 4.5.3.5.1.8, e.).
F = Flask internal pressure, (bar). (See 4.5.3.5.1.8, j.).
G = Atmospheric pressure, (bar). (See 4.5.3.5.1.8, j.).
H = Average acetylene peak areas. (See 4.5.3.5.1.8, l.).
T = Average ambient temperature, (Kelvin).
y = Mole percentage of acetylene in the reference gas, (See 4.5.3.5.1.9).

Besides magnesium acetylide, the expression can be used to find the amount of carbides, in terms of magnesium acetylide equivalent, that yield methane, ethene, ethane and other light hydrocarbons; substitute the appropriate "y" for each type and multiply the end result by the ratio {number of carbons in the hydrocarbon to that in MgC_2 (two)}.

4.5.3.5.2 Carbides (alternate 1).

4.5.3.5.2.1 Apparatus. Apparatus similar to that shown in Figure 3 shall be used. There must be a source of inert gas such as nitrogen, argon, or carbon dioxide.

4.5.3.5.2.2 Reagent. Hydroxylamine hydrochloride reagent.

a. Dissolve 0.25 grams of gelatin in 100 milliliters of hot distilled water in a beaker.
b. Dilute gelatin solution to 500 milliliters and add 500 milliliters of 95 % ethanol (or isopropanol) and 1.25 grams of hydroxylamine hydrochloride.

4.5.3.5.2.3 Procedure.

a. Add an accurately weighed portion of 50 ± 0.1 grams of magnesium powder to the reaction flask (A).
b. Transfer 50 milligrams of cuprous chloride to the test tube (B).
c. Pass nitrogen (or other inert gas) through the apparatus to remove the air and continue the passage of nitrogen throughout the determination.
d. Disconnect the test tube and add into it 15 milliliters of concentrated NH_4OH and 30 milliliters of the hydroxylamine hydrochloride reagent.
e. Reconnect the test tube.

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f. Add 100 milliliters of distilled water from dropping funnel (C) and switch on the heater.
g. Heat the mixture in the reaction flask just to boiling.
h. A red or pink coloration of the cuprous ammonium chloride solution in the test tube indicates that the magnesium sample contains more than 0.004 percent carbides. Should the cuprous ammonium solution become blue due to the introduction of air, it must be decolorized with successive 5 milliliter portions of hydroxylamine hydrochloride reagent until the blue color disappears.

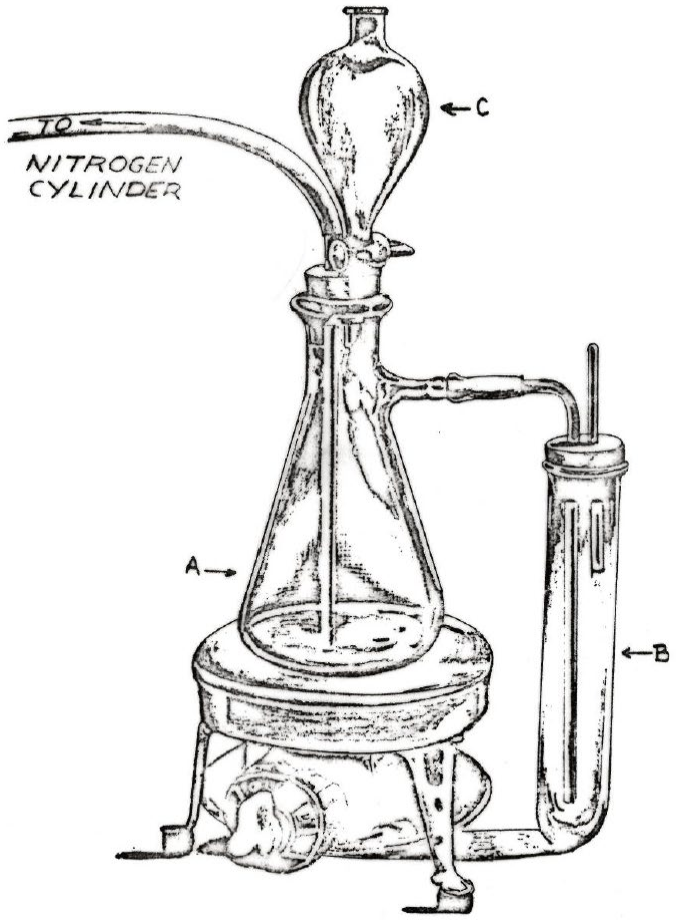


FIGURE 3. Apparatus for determination of carbides in magnesium.

4.5.4 Granulation. Place a weighed portioned of approximately 50 grams of the sample on the top sieve of a nest of sieves assembled as specified in Figure 1 and provided with a bottom pan. Cover and shake for 10 minutes in a mechanical shaker geared to produce 268 to 315 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portions retained by each sieve and calculate to a percentage as required (see 6.8).

4.5.5 Apparent density. Determine the apparent density of the magnesium powder, as specified in Table III, in accordance with the method described in ASTM B329.

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4.5.6 Workmanship. The magnesium powder shall be visually examined with unaided eyes by personnel with normal vision or normal corrected vision. The magnesium powder shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter or other defects that would render the magnesium powder unsuitable for the intended use.

4.5.7 Process verification.

4.5.7.1 Type I and II. The manufacturing process shall be demonstrated to be manufactured by grinding, stamping, or turning.

4.5.7.2 Type III. The manufacturing process shall be demonstrated to be manufactured use the atomizing process.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Magnesium powder covered by this specification is intended for use in the manufacture of tracer, igniter, signal flare and pyrotechnic compositions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Types and granulations required (see 3.3 and 3.5).

c. Whether a first article sample is required (see 3.1.1).

d. Packaging requirements (see 5.1).

6.3 Sample analysis. The instrument door should remain closed as much as possible during the analysis. Additional instructions may be found in the manufacturer's instruction manual.

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6.4 Prior type II granulation 16 requirements. MIL-M-382C, Amendment 1 included particle size requirements for Type II, Granulation 16 utilizing Sediograph 5000D which was subsequently deleted in MIL-DTL-382D. Due to extremely limited applicability, in combination with obsolescence of Sediograph 5000D model, and evolution of newer equipment and techniques, it has not been reintroduced in MIL-DTL-382E. (See 6.8 for additional information on newer equipment and techniques). For the extremely few programs which may need such additional requirements above the sieve granulation alone, they are encouraged to incorporate their program specific requirements directly in their call-out for magnesium within their Technical Data Package (drawings). Bearing in mind the above caveats, the historical call out would be specified as:

"Magnesium powder will be in accordance with MIL-DTL-382, Type II, Granulation 16; additionally, particle size will meet requirements of Table VII as assessed with a Sediograph 5000D (by Micromeritics) or equivalent (see, ASTM B761 or ISO 10076 (x-ray portion) or ISO 13317-3).

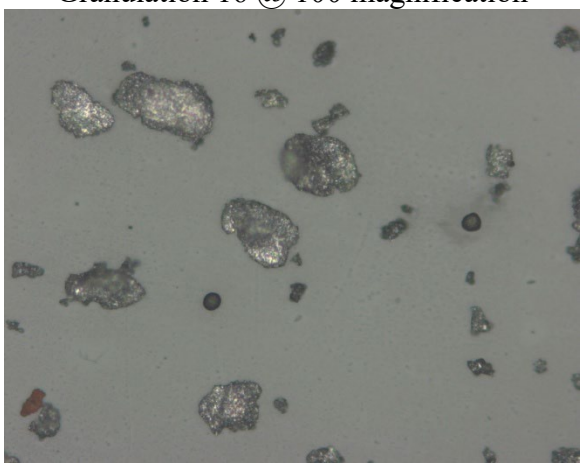
TABLE VII. Gravimetric particle size requirements.

Particle Ranges, microns		Percent %	
Greater than or equal to	But less than	Minimum	Maximum
0	4	0	0
4	14	8.4	14.7
14	30	21.6	27.5
30	74	58.3	66.6
74	-	0	4.0"

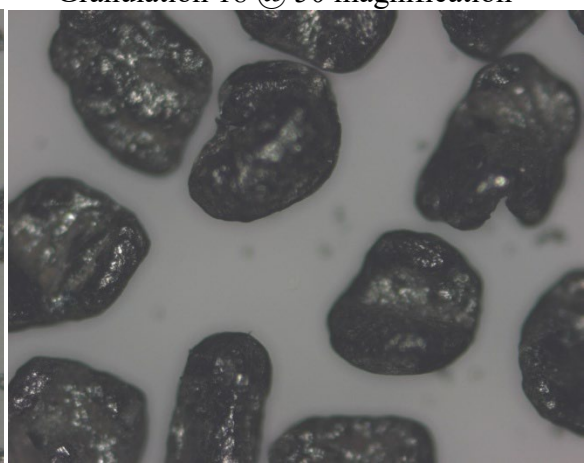
6.5 Types of Magnesium (examples).

Type I

Granulation 16 @ 100 magnification



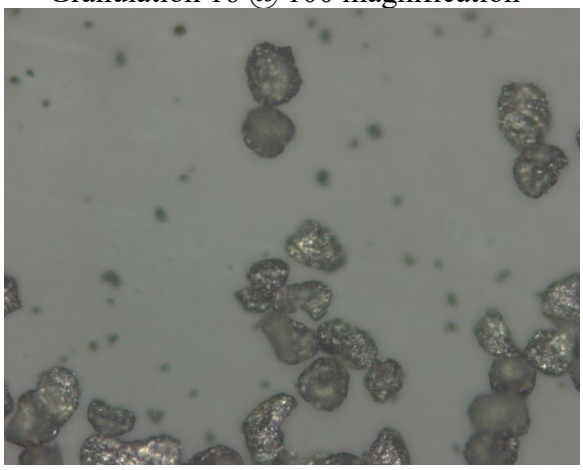
Granulation 18 @ 50 magnification



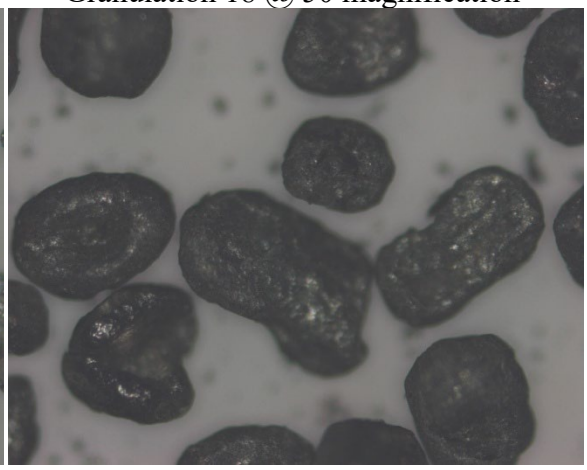
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Type II

Granulation 16 @ 100 magnification

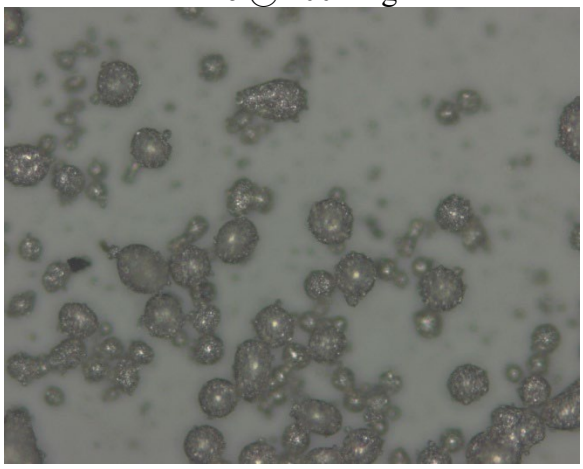


Granulation 18 @ 50 magnification

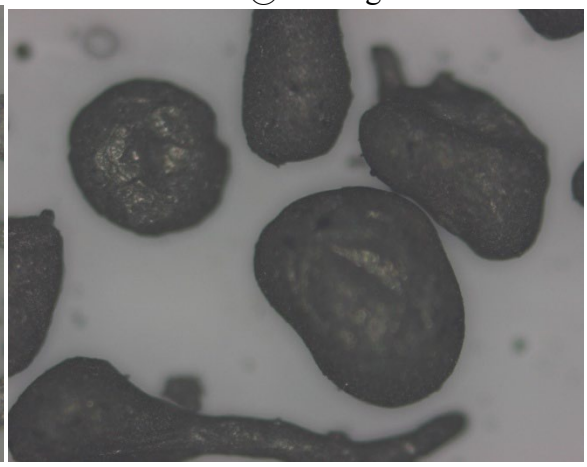


Type III

Granulation 16 @ 100 magnification



Granulation 18 @ 50 magnification



6.6 Units.

µm = micrometer
µg = microgram
g = gram
mg = milligram
mL = milliliter
min = minute
psi = pounds per square inch
kPa = kilopascal
m = meter
v/v = volume over volume
w/w = weight over weight
min = minimum
max = maximum

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6.7 Additional particle size requirements and analysis. Programs and suppliers may consider it desirable to impose additional requirements on particle size above and beyond those within the tolerances specified so as to specify further refinement of requirements or receive product optimized for their manufacturing processes. Furthermore, they may consider using alternate or additional methods and equipment to specify, characterize, and analyze size. Such analysis techniques may include:

a. Gravitational sedimentation (See ASTM B761 and ISO 10076)
b. Microscopy (see ISO 13322 and BS 3406-4)
c. Laser diffraction (see ASTM B822 and ISO 13320)

Those who are interested in alternate characterization techniques are cautioned that analysis and results from one technique are generally not directly comparable to other techniques due to numerous factors including particle morphology, measurement technique, etc. Beneficial information may be found on alternate techniques from the National Institute of Standards and Technology (NIST) Special Publication 960-1, Recommended Practice Guide on Particle Size Characterization, available at http://www.nist.gov.

6.8 Changes incorporated. This specification was updated from MIL-DTL-382D to MIL-DTL-382E with Amendment 2 as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Details of these changes are documented in the ECPs and ERR listed below.

Specification ID	ECP / ERR
MIL-M-382C	-
MIL-M-382C w/ Amendment 1	A1D2001
MIL-DTL-382D	-
MIL -DTL-382E	R13Q2080
MIL-DTL-382E w/Amendment 1	R15Q2008
MIL-DTL-382E w/Amendment 2	Y24Q3003

6.9 Subject term (key word) listing.

Illumination flare
Igniter
Pyrotechnic
Signal flare
Tracer
Countermeasure flare
Munitions Chemical Specification Modernization (MCSM)

6.10 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

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Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodian:
Army- AR
Navy- OS
Air Force- 11

Preparing activity:
Army- AR
(Project: 6810-2025-006)

Review Activities:
Army- EA, MI, MR, TE
Navy- AS
Air Force- 03, 68

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